Normalized Conditional Transports: Learning and Execution Checks

Research note with independent mathematical checks

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Abstract

We give an invertible realization of an existing scale mixture, an output-energy exclusion bound, and a learned conditional-CDF transport with explicit finite-sample KL risk under bounded local conditional complexity. All constructions retain a full-dimensional Gaussian source. Independent checks cover inverse maps, Jacobians, and the statistical unit of observation. A real-data covariance experiment improves its parent but fails against block and Student controls. These results establish restricted mathematical and numerical properties. They do not establish superior image generation, video generation, efficiency, or representation relative to latent diffusion.

1 An invertible realization of the existing mixture

At a fixed condition, let the desired three-dimensional density be

$$q(x) = \sum_{k=1}^{K} w_k \mathcal{N}_3(x; 0, s_k^2 L L^\top), \tag{1}$$

where K is the number of mixture components, $w_k > 0$ sum to one, $s_k > 0$ are scales, and L is an invertible lower-triangular matrix with positive diagonal. The implementation uses $\det L = 1$. Let F be the standard three-dimensional Gaussian radius CDF and define $H(r) = \sum_k w_k F(r/s_k)$ for $r \geq 0$. Both CDFs are strictly increasing at positive radii.

Proposition 1. *For $u \in \mathbb{R}^3$ with $\rho = \|u\| > 0$, set $r = H^{-1}(F(\rho))$ and*

$$T(u) = L\frac{r}{\rho}u, \qquad T(0) = 0. \tag{2}$$

The exact map is an invertible transport from a standard Gaussian to q. Its inverse whitens x by L^{-1} and maps radius r to $F^{-1}(H(r))$.

For an RGB detail block, the map retains all three source coordinates and represents the same mixture density as the legacy component sampler. It introduces numerical inverse-CDF work. The legacy sampler remains a distributional control, but overlapping components prevent recovering its input uniquely from its output.

Proof. A standard Gaussian has independent uniform direction and radius CDF F. The radial quantile map changes its radius CDF to H and preserves that direction. Multiplication by L gives the stated density. Strict monotonicity of the radial map and invertibility of L prove bijectivity. At a positive radius, its determinant is

$$|\det DT(u)| = \det L \left(\frac{r}{\rho}\right)^2 \frac{F'(\rho)}{H'(r)}. \tag{3}$$

Consequently its log determinant is $\log \phi_3(u) - \log q(T(u))$, where ϕ_3 is standard Gaussian density. At zero the radial scaling tends to $(\sum_k w_k s_k^{-3})^{-1/3}$, since both radial CDFs have cubic leading terms. The ratio is a smooth function of squared radius by the implicit function theorem applied to the positive cubic-leading CDF expansions, giving the smooth extension at the origin. $\square$

The root satisfies $s_{\min}\rho \le r \le s_{\max}\rho$, where $s_{\min}$ and $s_{\max}$ are the smallest and largest component scales. Bisection to relative tolerance ϵ takes order $\log((s_{\max}/s_{\min} - 1)/\epsilon)$ iterations for unequal scales. Each forward iteration evaluates K component probabilities. The code uses the smaller tail in logarithmic form and raises on nonconvergence. Its reported brackets rely on floating-point special functions; they are not interval-arithmetic proofs. The density-ratio log determinant approximates the derivative of the exact transport, with error from root inversion.

2 Rejecting corrections too small for the output target

Let P be a target law on $\mathbb{R}^D$ and Q a generated law, both with finite first moments. Define the energy score

$$\mathrm{ES}(Q; P) = \mathbb{E}\|X - Y\| - \tfrac{1}{2}\mathbb{E}\|X - X'\|, \tag{4}$$

where X, X' are independent with law Q, and Y independently has law P.

Proposition 2. *Let (X, Z) be any coupling of generated laws Q, Q' with finite first moments. Then*

$$|\,\mathrm{ES}(Q; P) - \mathrm{ES}(Q'; P)| \le 2\mathbb{E}\|X - Z\|. \tag{5}$$

The constant two is sharp uniformly over these laws.

A small paired output displacement rules out a large improvement against every target with a defined score. This check uses only generated samples. A large displacement gives no evidence of improvement. The result does not apply to FID or establish downstream prediction accuracy.

Proof. The reverse triangle inequality bounds the change in the target-distance term by $\mathbb{E}\|X - Z\|$. Apply the same inequality to an independent copy (X', Z'). The change in the self-distance term is bounded by $\frac{1}{2}\mathbb{E}(\|X - Z\| + \|X' - Z'\|)$, giving the result. For sharpness, take P as a point mass at zero, Q as a point mass at one, and Q' as masses $1 - a, a$ at one and zero, where $0 < a < 1$. The score change is $2a - a^2$ and the optimal mean displacement is a. Their ratio approaches two as a tends to zero. $\square$

For normalized image distance, divide distances and score margins by $\sqrt{D}$. Freeze both generators before drawing n independent paired sources. Suppose each normalized displacement d_i has a known almost-sure bound b. For M prespecified comparisons and error probability $0 < \alpha < 1$, Hoeffding's inequality and a union bound give the simultaneous upper bound

$$U = \min\left\{ b, \frac{1}{n}\sum_{i=1}^{n} d_i + b\sqrt{\frac{\log(M/\alpha)}{2n}} \right\}. \tag{6}$$

Reject a requested positive score margin η only when $2U < \eta$. The bound b must follow from the generators or declared preprocessing. An observed maximum is insufficient. Candidate adaptation requires fresh independent diagnostic sources and an error allocation across rounds. For paired finite arrays and the usual unbiased self-distance estimate, the score change is deterministically bounded by twice the sample mean displacement; this finite-array fact alone supplies no population guarantee.

3 A smooth nonlinear-generated dilution example

Let A be uniform on the four planar unit-axis vectors, and let C have the law of A rotated by $\pi/4$. Add independent Gaussian smoothing with variance $\tau^2 > 0$ in each active coordinate. Append m independent Gaussian coordinates with variance $\sigma^2 > 0$, keeping τ, σ fixed as m grows. The resulting laws are smooth, positive, full-dimensional, and non-Gaussian. Each active law can be generated by a nonlinear triangular Gaussian quantile transport.

The excess energy score between the rotated and unrotated laws satisfies

$$\Delta_m \sim \frac{5}{32}\sigma^{-7}2^{-7/2}m^{-7/2}. \tag{7}$$

Normalization by $\sqrt{m+2}$ gives order $(m+2)^{-4}$. To verify the coefficient, the unsmoothed cross/self squared distances have equal moments through order three and a fourth-moment difference of -4. For complement distance $r > 0$ and active squared distance $v \geq 0$, the fourth Taylor coefficient of $\sqrt{r^2 + v}$ is $-5/(128r^7)$. Gaussian smoothing preserves the cancellations. For $m > 9$, a fifth-order remainder is integrable and smaller by order m^{-1}. The complete moment table, remainder, inverse-chi calculation, and independent check accompany this note.

This example explains how a real active-coordinate improvement can become small in a global score. It does not establish that the previous image experiments have its symmetry or decay rate. It motivates measuring output effects directly before promoting a latent dependence result.

4 A learned conditional transport with a finite risk bound

The radial construction fixes an endpoint law. We now give a different estimator whose endpoint is learned from observations. This estimator supplies a checked learning guarantee; it does not justify the optimization of the separate neural spline model.

Let $X^{(1)}, \ldots, X^{(n)}$ be independent complete observations. Each image or video clip is one statistical unit. A fixed C^1 diffeomorphism maps observations to $Y \in (0,1)^D$. Fix an ordered observed-coordinate graph, with context C_j containing $k_j \leq k$ predecessors of coordinate j. Assume

$$p(y) = \prod_{j=1}^{D} p_j(y_j \mid c_j), \qquad a \leq p_j(t \mid c) \leq A, \tag{8}$$

where $0 < a \leq 1 \leq A < \infty$. For $0 < \beta \leq 1$, assume

$$|p_j(t \mid c) - p_j(t' \mid c')| \leq L\|(t,c) - (t',c')\|^{\beta}. \tag{9}$$

The chart and graph must be fixed independently of these observations, or their learning requires a separate argument. No independence among coordinates is assumed.

Divide each response and context axis into b equal bins. For a context cell l, let N_l count its observed clusters and N_{lr} count responses in bin r. Fit

$$\widehat{\pi}_{lr} = \frac{N_{lr} + 1}{N_l + b}. \tag{10}$$

At an observed context c, interpolate the response densities at neighboring context-cell centers with nonnegative multilinear weights summing to one. Clamp contexts to the nearest center at the boundary. The resulting density $\widetilde{q}_j(t \mid c)$ is positive and normalized, continuous in c, and constant within each response bin.

Proposition 3. *Suppose the marginal density of C_j is bounded below by $a_{c,j} > 0$. The interpolated estimator $\widetilde{Q} = \prod_j \widetilde{q}_j$ satisfies*

$$\mathbb{E}_{\text{train}} \operatorname{KL}(P\|\widetilde{Q}) \leq \sum_{j=1}^{D} \left\{ \frac{L^2(1+9k_j/4)^\beta}{a} b^{-2\beta} + \frac{b^{k_j}(b-1)}{(n+1)a_{c,j}} \right\}. \tag{11}$$

Under the factorization and lower conditional bound above, $a_{c,j} = a^{k_j}$ is valid. For bounded constants and fixed k, choosing b of order $n^{1/(2\beta+k+1)}$ gives expected KL per coordinate of order $n^{-2\beta/(2\beta+k+1)}$.

This bound derives fitting and approximation error from observed sample size and regularity. Total joint KL retains its factor D. A small per-coordinate error alone does not establish a small full-image error.

Proof. For a fixed context cell with s observations and true response-bin probabilities π_r, the binomial reciprocal identity gives

$$\mathbb{E}\frac{1}{N_r+1} \leq \frac{1}{(s+1)\pi_r}. \tag{12}$$

Jensen's inequality bounds each category's expected log ratio by $\log((s+b)/(s+1)) \leq (b-1)/(s+1)$. The bound is independent of the category, so it also holds when weighted by the response probabilities at a neighboring query context. If a cell has target probability w_l, then

$$\mathbb{E}\frac{1}{N_l+1} \leq \frac{1}{(n+1)w_l} \leq \frac{b^{k_j}}{(n+1)a_{c,j}}. \tag{13}$$

Every point of an interpolating context cell lies within $3/(2b)$ per context axis of the query. Response-bin distance is bounded by $1/b$. The Hölder condition and $\operatorname{KL}(p\|q) \leq \int (p-q)^2/q$ bound approximation error by the first term. Convexity of KL in its second density permits averaging over interpolation weights. Sum the conditional bounds by the joint KL chain rule. Finally, each selected-coordinate density conditional on earlier selected coordinates averages a full preceding-coordinate conditional bounded below by a, giving the context lower bound a^{k_j}. □

The complete sampling algorithm takes $Z \sim \mathcal{N}_D(0, I)$ and, in graph order, sets

$$Y_j = \widetilde{F}_j^{-1}(\Phi(Z_j) \mid C_j), \tag{14}$$

where Φ is the standard Gaussian CDF and $\widetilde{F}_j$ integrates the fitted conditional density. Each CDF is continuous and strictly increasing. Its inverse is piecewise linear in the response probability. The complete triangular map is continuous and invertible, with the change-of-variables density almost everywhere. It is not generally globally C^1 at knots. All D source coordinates are retained.

Tables require order Db^{k+1} storage. Counting costs order $nD(k+1)$ plus table construction. Generation by interpolated-CDF binary search costs order $D2^k(k + \log b)$, plus the observation chart. Graph depth controls sequential sampling: a chain has depth D, while a balanced bounded-branching tree has depth of order $\log D$. These operation counts include no measured advantage over a neural decoder.

For a concrete nonlinear, non-Gaussian example, take a uniform root and tree conditionals

$$p_j(t \mid c) = 1 + \theta \cos(2\pi c)\cos(2\pi t), \qquad |\theta| < 1. \tag{15}$$

Their strictly increasing conditional CDFs are $t + \theta\cos(2\pi c)\sin(2\pi t)/(2\pi)$. The theorem gives expected KL per coordinate of order $n^{-1/2}$ for this one-parent Lipschitz model. The implementation supports zero or one parent, matching this example.

The omitted-context failure is equally explicit. For

$$p_\theta(y) = 1 + \theta \cos(2\pi y_1) \cos(2\pi y_D), \tag{16}$$

any local predecessor graph that omits y_1 from the final conditional has uniform true local conditionals and an irreducible joint KL of at least $\theta^2/[8(1+|\theta|)]$. The complete derivation and finite histogram-comparison constants are in the accompanying mathematical note. Those comparisons concern a specified estimator; they do not prove a separation from trained latent diffusion.

A stochastic latent decoder can use the same fitted tables, graph, and Gaussian inputs and produce exactly the same distribution and computation. The implementation tests this equality. A strict advantage against every such decoder is impossible. Invertibility also gives no general semantic or equal-dimensional representation advantage over a VAE.

5 Smooth convex heads and sharper estimation error

A second estimator uses positive continuous response densities and continuously differentiable context features. This is the current convex implementation; its fitting guarantee is distinct from the histogram theorem and from neural flow optimization. Let fixed causal feature weights satisfy

$$w_{jr}(y_{<j}) \geq 0, \qquad \sum_{r=1}^{R_j} w_{jr}(y_{<j}) = 1. \tag{17}$$

Let H_k be linear response hats at k/b, with integral weights $v_0 = v_b = 1/(2b)$ and $v_k = 1/b$ otherwise. Define

$$q_{a,j}(t \mid y_{<j}) = \sum_r w_{jr}(y_{<j}) \sum_{k=0}^{b} a_{g(j),r,k} H_k(t), \quad \ell \leq a_{g,r,k} \leq U, \quad \sum_k v_k a_{g,r,k} = 1. \tag{18}$$

Here $0 < \ell < 1 < U$. The feasible coefficient set is convex, every conditional density is positive and normalized, and the complete map $Y_j = F_{a,j}^{-1}(\Phi(Z_j) \mid Y_{<j})$ retains all D Gaussian inputs. Within a response interval the density is linear, so its CDF is quadratic and has a stable scalar inverse. With continuously differentiable causal weights the exact transport is continuously differentiable. The implemented weights are quadratic tensor-product B-splines of zero to two preceding coordinates. Neural feature weights are a prospective extension, with their learning cost and approximation error unresolved.

Fit the coefficient vector by normalized full-array negative log likelihood. Independent arrays are the sampling units even when sites share parameters. The Frank–Wolfe gap at a feasible returned vector bounds its empirical optimization error. Failure to reach a requested gap is recorded explicitly.

Proposition 4. *Fix the features, graph, groups, and observation chart independently of n fitting arrays. Assume the target has finite KL to one class member. Let P_a count all separately fitted coefficients, let a_* minimize population loss over the actual convex class, and let $\widehat{a}$ have empirical suboptimality bounded by τ. Put*

$$B = \log(U/\ell), \qquad V_B = \frac{B^2}{B - 1 + e^{-B}}, \qquad t_n = P_a \log\left(1 + \frac{8(U-\ell)n}{\ell}\right) + \log(1/\delta). \tag{19}$$

With probability at least $1-\delta$,

$$\frac{\mathrm{KL}(P \| Q_{\widehat{a}}) - \mathrm{KL}(P \| Q_{a_*})}{D} \leq \left(2V_B + \frac{4B}{3}\right) \frac{t_n}{n} + 2\tau + \frac{1}{n}. \tag{20}$$

The claim holds under misspecification.

This bound controls additional loss incurred by fitting coefficients. The remaining approximation loss depends on frozen features and selected contexts.

Proof. Write

$$r_a(X) = D^{-1} \sum_j \log(q_{a*,j}/q_{a,j}).$$

Population first-order optimality implies

$$\mathbb{E}\left[D^{-1} \sum_j q_{a,j}/q_{a*,j}\right] \leq 1.$$

The arithmetic–geometric mean inequality within each array gives $\mathbb{E}e^{-r_a} \leq 1$, without coordinate independence. For $|r| \leq B$, Taylor's integral identity gives

$$e^{-r} - 1 + r = r^2 \int_0^1 (1-u)e^{-ur}\, du \geq \frac{B - 1 + e^{-B}}{B^2} r^2. \tag{21}$$

The resulting moment bound is $\mathbb{E}r_a^2 \leq V_B \mathbb{E}r_a$. One-sided Bernstein concentration and Young's inequality yield, for each fixed feasible net point, $\mathbb{E}r_a \leq 2P_n r_a + (2V_B + 4B/3)t_n/n$ on the simultaneous net event. A feasible coefficient net of radius $\ell/(4n)$ has the stated logarithmic size; the partition of unity transfers this radius to density, and log-loss discrepancy is bounded by $1/(4n)$. Lift the empirical τ-optimizer to that net and transfer back. The resulting remainder $3/(4n)$ is bounded by $1/n$. □

The theorem derives estimation error for frozen features. It does not prove that a learned neural analysis discovers sufficient image context. If discovery arrays learn a chart or features, freeze them before independent head fitting; the conditional theorem then applies to the heads alone. Neither reused development data nor a new split of previously studied data supplies fresh campaign-level confirmation.

For the cosine example with parameter 0.65, the sharper coefficient is 12.5849, compared with 5697.3011 from the earlier curvature bound. Even with zero optimization error, the headwise bound with simultaneous coverage is 2.8587 nats per coordinate at 40,000 arrays after integer mesh search. The restricted independent-coordinate lower bound is 0.01600. These sufficient bounds do not establish a practical separation. A global stochastic latent decoder can still copy the full construction exactly.

6 Executed checks and remaining work

The full QALT suite passed 208 tests after adding complete generators, the learned-analysis latent baseline, global conditional residual couplings, strict observed-data preparation, and feasible acceleration. Sixteen separate deterministic checks of the sharper proof and resource calculations passed. These checks include normalization, dense numerical Jacobians, frozen-analysis gradients, causal generation, and independent constrained optimization references. The earlier PSC numerical fixture had nonlinear six-dimensional inverse error about 2.2×10^{-12}. Its deliberately tiny correction had a normalized score-change upper bound 0.000185, below the declared margin 0.005. The radial inverse used 44 CDF iterations and was substantially slower than the legacy mixture sampler.

PSC job 45550810 completed the frozen covariance experiment on 5,000 reused CIFAR development images. Every fitted arm used the same training coordinates, and the official test batch remained unopened. Positive values in Table 1 favor full covariance. The block and Student requirements failed. The descriptive intervals do not account for the earlier adaptive model search.

Comparator	Gain (nats/detail)	Required gain	Decision
B4 parent	0.3184877	> 0.01	Pass
Block covariance	−0.0000028	> 0.01	Fail
Fixed Student	−0.1834003	> -0.01	Fail

Table 1: Completed adaptive-development covariance comparison. Full covariance adds no material benefit over block covariance and remains below Student.

The global Gaussian optimum on the same development arrays gains only 0.00009782 nats/detail above fitted full covariance. It cannot close the observed Student deficit within that model class on these arrays. Its calculation supplies no population coverage and selects no fitted model. Source, score, and result hashes were verified independently.

PSC job 45552606 completed all 36 prescribed conditional-CDF cells. For the nonlinear tree with $D = 8$, mean estimated KL per coordinate across three fits decreased from 0.03028 to 0.01525 to 0.00744 at training sizes 256, 1,024, and 4,096; the corresponding $D = 32$ values were 0.03041, 0.01567, and 0.00762. These are descriptive estimates, with no empirical asymptotic-rate claim. The distant-dependence study remained above its joint lower bound 0.03201; no plateau was established. All 38 result hashes and seven source hashes were independently checked. Source round-trip error was below 1.1×10^{-12}, and all 36 copied stochastic decoders returned bitwise-identical outputs. Runner time was 9.74 seconds, with no competing-method timing claim.

PSC job 45554441 completed 36 positive-spline cases, but only 14 reached the required 10^{-4} optimization gap. All 18 favorable local fits and four distant fits exhausted 500 updates. Local $D = 32$ mean KL per coordinate decreased from 0.001397 at 256 arrays to 0.000547 at 4,096 arrays. The distant generated first/last cosine moment remained near zero, against the target 0.1625. All numerical and copied-decoder checks passed. Independent reconstruction of saved coefficients and the published evaluation streams reproduced KL, energy, and moments within 3.4×10^{-16}, without refitting. It leaves all optimization and dependence failures intact.

The L40S job 45552255 was canceled before execution after its estimated wait reached five days. Replacement GPU job 45554440 failed before training with a CUDA driver/runtime mismatch. A separate diagnostic passed on another node using the unchanged runtime. The one-case CPU pilot 45558401 and four-case GPU pilot 45558567 then completed. Independent recomputation reproduced all 150 saved metrics within 8.4×10^{-9}. All recorded numerical checks and exact copied-decoder equalities passed. Quality rankings varied by case. In the GPU records, the spline used more allocated training memory than both FM controls and was slower than several FM configurations. First-call timing effects prevent a reliable speed frontier. The four-channel cases contain synthetic tensors. They provide no real-video evidence.

7 Current candidate: learned analysis and global residual transport

For native RGB 32×32 arrays, retain all $D = 3072$ coordinates. Apply the same logit in float64 before conversion to float32. A learned invertible pre-Haar map followed by a multiscale spline analysis gives $A(y) = (c, r)$, with $d = 192$ coarse coordinates and $D - d = 2880$ residual coordinates. Lossless packing places the residuals on the 8×8 coarse grid with 45 channels. The learned pre-Haar map permits fine input information to change the retained code. Invertibility alone does not establish a representation with improved semantic prediction.

Train A by full Gaussian likelihood, then freeze it. Fit a shared coarse FM and compare a four-layer global conditional coupling decoder with a globally attending conditional residual FM. For a fixed subset J in one coupling layer,

$$r'_J = r_J, \qquad v_{J^c} = e^{s(r_J, c)} \odot r_{J^c} + t(r_J, c), \tag{22}$$

$$r'_{J^c} = \mathrm{RQS}\{v_{J^c}; h(r_J, c)\}. \tag{23}$$

The attention conditioner sees the entire fixed residual subset and coarse grid. Bounded log scales, positive spline derivatives and linear tails make each conditional coordinate map a real-line bijection. Alternating masks permit nonlinear dependence across scales and distant positions.

Proposition 5. *For any fixed coarse vector c, the composed residual map T_c is invertible with normalized conditional density*

$$q_R(r \mid c) = \phi_{D-d}(T_c^{-1}(r))\,|\det D_r T_c^{-1}(r)|. \tag{24}$$

For any coarse law, independent residual Gaussian noise and inverse analysis define a normalized observation law. A coarse density yields a D-dimensional observation density.

This retains every source coordinate and gives an explicit conditional density. The proposition imposes no conditional independence among residual coordinates.

Proof. Order fixed coordinates before transformed coordinates. Each layer Jacobian is triangular, with positive diagonal entries equal to the affine scale times the spline derivative. The fixed coordinates recover the conditioner parameters in the inverse. Reverse layer order to invert the composition. Conditional change of variables gives integral one. Integrating this conditional law against the coarse law, then applying inverse analysis and sigmoid, preserves total mass. □

A density formula for the complete model additionally requires a coarse density. The implemented finite-step FM law is evaluated using its actual samples. Its training MSE and the separate analysis likelihood are not reported as complete-model NLL. A stochastic latent decoder copying every map, weight and source draw ties the candidate exactly.

The native CIFAR input check 45558402 passed source and numerical checks for 4,000 fitting and 1,000 reused development arrays; the official test remained unopened. Pilot 45567461 allocates 90 seconds to analysis, 90 to shared coarse fitting, and 180 per residual arm. Residual parameter counts are 261,524 for the spline and 263,421 for FM, fixed by shape before fitting. The full Gaussian source and residual-coordinate ordering are identical. Residual FM uses 4, 8, 16, 32 and 64 velocity evaluations. An analysis-only Gaussian control tests whether additional modeling improves the fitted analysis.

The study saves generated pairs, checks actual sources and record identities, and charges full fitting and generation costs. Reused development data provide no fresh confirmation. Fewer flow layers do not ensure lower cost. Neither proposition bounds learned-analysis approximation, estimation, optimization or numerical error at practical budgets. Real-image and real-video superiority, representation probes, stronger budget-matched controls and untouched confirmation remain unresolved.